THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS TWO-CHAPTER TEST SET D

Chapters 6 & 10: **Triangles, Circles** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. $\triangle PQR \sim \triangle XYZ$ with $PQ = 3$ cm, $XY = 6$ cm and $YZ = 10$ cm. Then QR is: 1

(a) 20 cm **(b)** 6 cm **(c)** 5 cm **(d)** 4 cm

2. In $\triangle ABC$, $DE \parallel BC$, $\frac{AE}{EC} = \frac{1}{2}$ and $AD = 3$ cm. Then DB is: 1

(a) 3 cm **(b)** 1.5 cm **(c)** 6 cm **(d)** 9 cm

3. Which of the following is **not** a similarity criterion? 1

(a) SAS **(b)** AA **(c)** RHS **(d)** SSS

4. A student states: “Two congruent triangles are also similar.” The correct response is: 1

(a) Correct only for isosceles triangles **(b)** Correct

(c) Incorrect **(d)** Cannot be determined

5. $\triangle ABC \sim \triangle DEF$ with $\frac{AB}{DE} = \frac{1}{2}$ and the perimeter of $\triangle ABC$ 15 cm. Student A says the perimeter of $\triangle DEF$ is 30 cm; Student B says 7.5 cm. Then: 1

(a) Student A is correct **(b)** Student B is correct

(c) Both are correct **(d)** Both are wrong

6. If OP is a radius and PT the tangent at P, then $\angle OPT$ is: 1

(a) 45° **(b)** 60° **(c)** 90° **(d)** 180°

7. The tangent from a point 10 cm from the centre is 8 cm long. The radius is: 1

(a) 6 cm **(b)** $\sqrt{164}$ cm **(c)** 2 cm **(d)** 18 cm

8. Tangents PA and PB make $\angle APB = 120^\circ$. Then $\angle AOB$ is: 1

(a) 240° **(b)** 60° **(c)** 120° **(d)** 30°

9. A student states: “From a point on a circle, two tangents can be drawn.” The correct response is: 1

(a) Correct

(b) Infinitely many

(c) None can be drawn

(d) Incorrect — exactly one tangent can be drawn

10. Two concentric circles have radii 10 cm and 8 cm. For the chord of the larger circle touching the smaller one, Student A says 12 cm; Student B says 6 cm. Then: 1

(a) Student A is correct **(b)** Student B is correct

(c) Both are correct **(d)** Both are wrong

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. In $\triangle ABC$, $DE \parallel BC$ with $AD = 3$ cm, $AB = 7.5$ cm and $EC = 3$ cm. Find AE. 2

12. $\triangle ABC \sim \triangle PQR$ with $\angle A = 32^\circ$ and $\angle R = 65^\circ$. Find $\angle B$. 2

13. A tangent PQ at P of a circle of radius 9 cm meets a line through the centre at Q with $OQ = 41$ cm. Find PQ. 2

14. The incircle of $\triangle ABC$ touches AB at F, BC at D and CA at E. If $AF = 4$ cm, $BD = 3$ cm and $CE = 5$ cm, find the perimeter of the triangle. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In $\triangle ABC$, D and E lie on AB and AC with $AD = 1.5$, $BD = 3$, $AE = 1$ and $CE = 2$ cm. Prove that $DE \parallel BC$ and find DE if $BC = 6$ cm. 3

16. Prove that the tangent drawn at the mid-point of an arc of a circle is parallel to the chord joining the end points of the arc. 3

17. $\triangle ABC \sim \triangle DEF$ with $AB = 4$, $BC = 6$, $CA = 8$ cm and $DE = 6$ cm. Find EF and FD, and the ratio of their perimeters. 3

OR

The diagonals of trapezium ABCD ($AB \parallel DC$) meet at O. If $OA = 6$ cm, $OC = 8$ cm and $OB = 3$ cm, find OD. 3

18. A circle is inscribed in $\triangle ABC$ with $AB = 8$ cm, $BC = 10$ cm and $CA = 12$ cm, touching AB at F, BC at D and CA at E. Find AF, BD and CE. 3

OR

PQ is a chord of length 4.8 cm of a circle of radius 3 cm. The tangents at P and Q meet at T. Find TP. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** A map of a triangular park ABC is drawn to scale with sides 3 cm, 4 cm and 5 cm. The shortest side of the actual park is 60 m. 5

(i) Find the scale (metres per cm).

(ii) Find the other actual sides.

(iii) Find the cost of fencing the park at ₹50 per metre.

OR

In $\triangle ABC$, P and Q are points on AB and AC such that $PQ \parallel BC$. Prove that the median AD bisects PQ. 5

20. **Case study:** A circular coin of radius 1.5 cm lies in the corner of a box, touching two perpendicular walls at A and B. Let C be the corner. 5

(i) Find CA and CB.

(ii) Find the distance of the centre O from C.

(iii) Which property makes CA = CB?

OR

Prove that the tangent at any point of a circle is perpendicular to the radius through the point of contact. Hence prove that the tangents at the ends of a diameter are parallel. 5

END OF QUESTION PAPER — Total: 40 Marks

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